

THE GAMMA POINT PROCESS

H. Linhart
University of Natal, Durban

1. Introduction

Stationary point processes have attracted the interest of statisticians for some time and much valuable work has been done to investigate their properties. There is, however, a considerable gap as far as models with a finite number of parameters are concerned. There is the thoroughly explored Poisson process, and there are other, more general, renewal processes, but there is little on processes for which the distances between points are not independently distributed. There is work on the problems of estimation for the general stationary point process (Bartlett (1963a, 1963b), Cox (1965)) but I do not know of work on statistical problems for any model with a finite number of parameters, apart from the special cases mentioned.

In this paper a process is studied for which any finite number of distances between consecutive points have a multivariate gamma distribution. The choice is, of course, arbitrary because any multivariate distribution of non-negative variables can be used to define a stationary point process. But the gamma distribution has proved to be very prevalent in univariate situations with non-negative variables and it seems natural to expect something similar in the multivariate case. The gamma point process includes as special cases, the gamma renewal process (Erlang process) and the Poisson process. The aim of this paper is to study the properties of this process and to provide statistical methods which can be used if one feels justified to postulate a gamma point process.

2. Remarks on terminology

I shall not speak about *events* (e.g. numbers of events in certain intervals) but about *points*, mainly because the processes are called point processes. Nevertheless, I shall continue to call the basic underlying variable *time*, in spite of the fact that some of the terms used refer to linear extent rather than to time. But time, points in time, distances between time points, lengths of intervals between time points, seems to be acceptable.

It is well known (cf. e.g. McFadden (1962) or Cox and Lewis (1966)), that two different discrete parameter stochastic processes arise in connection with point processes, namely, those of the distances between (or lengths of intervals between) points depending on whether "the origin is at an arbitrary event" or "the origin is at an arbitrary time" (Cox and Lewis (1966) p. 67). Cox denotes the first process X_i , i.e.f., where $i = \dots, -2, -1, 0, 1, \dots$, and the other L_i , i.e.f. I shall adopt his notation and call the X_i 's *distances* (distances between points) and the L_i 's *lengths* (lengths of intervals between points).

The *distances* can be used conveniently for the specification of a point process, but what one usually *observes* are the *lengths*. A rigorous definition of the processes of distances and lengths is mathematically intricate, as indeed is the definition of a point process. The basic definitions have been given by Ryll-Nardzewski (1961), they are discussed, together with problems of specification, by Steinijans (1970).

3. The multivariate gamma distribution and the gamma process

The multivariate gamma distribution (Krishnamoorthy and Parthasarathy (1951), Krishnaiah and Rao (1961)) has characteristic function

$$\varphi(u_1, u_2, \dots, u_n) = |I - i D(u_1, u_2, \dots, u_n) \Sigma_n|^{-\beta},$$

where $\beta > 0$, $D(u_1, u_2, \dots, u_n)$ is a diagonal matrix with elements $u_1, u_2, \dots, u_n$ and Σ_n a real, symmetric, positive definite $n \times n$ matrix.

We want to use this distribution to define a stationary continuous parameter stochastic process $X(t)$, $t \in R_1$. By Kolmogorov's Theorem (cf. e.g. Cramér and Leadbetter (1967)) it is sufficient for this to define the finite dimensional distributions of the process in such a way that the conditions of symmetry and consistency are satisfied. The condition of symmetry ($F(X(t_1), \dots, X(t_n)) (x_1, x_2, \dots, x_n)$ invariant under permutations of the pairs (x_i, t_i)) is trivially satisfied if one defines the joint distribution of $X(t_1), \dots, X(t_n)$ to be *some* multivariate distribution. The condition of consistency is satisfied if any r -dimensional marginal distribution of the n -dimensional distribution *postulated* for $X(t_1), X(t_2), \dots, X(t_n)$ ($r \leq n$; n and $t_1, \dots, t_n$ arbitrary) is identical to the corresponding *postulated* r -dimensional distribution. In our case this puts a condition on the parameter matrix Σ_n of the postulated distributions. Let $R^*(x, y)$ be any symmetric, positive definite, real valued function defined for all $x, y \in R_1$ and postulate for $X(t_1), \dots, X(t_n)$ (n and $t_1, t_2, \dots, t_n \in R_1$ arbitrary) the multivariate gamma distribution with parameters β and $\Sigma_n = \{R^*(t_i, t_j)\}$, then it is easily seen from the characteristic functions that the finite dimensional distributions are consistent. ($R^*(x, y)$ has to be real valued, symmetric, and positive definite for Σ_n to be real, symmetric and positive definite).

It can also be seen from the characteristic function that $X(t)$ will be strictly stationary if $R^*(x, y)$ depends only on $|x - y|$, thus if $R^*(x, y) = R(|x - y|)$. As we are only interested in stationary processes here, we shall assume that $R^*(x, y) = R(|x - y|)$ in all that follows.

We call the stochastic process with the above given finite dimensional distributions the *gamma process* with parameters β and $R(\tau)$, $\tau \geq 0$. It is easily verified from the characteristic function of the finite dimensional distributions that the process has mean $\beta R(0)$ and covariance function $\text{Cov}(X(t), X(t + \tau)) = \beta R^2(|\tau|)$. The probability density function of $X(t)$ is that of a gamma distribution with parameters $a = R(0)$ and β .

A gamma distribution with parameters a and β has density

$$f(x) = x^{\beta-1} e^{-x/a} / a^\beta \Gamma(\beta) \quad \begin{array}{l} x \geq 0, \\ a, \beta > 0, \end{array}$$

and characteristic function

$$\varphi(u) = (1 - i u a)^\beta;$$

its mean is $a\beta$ and its variance $a^2\beta$.

We give below a few moments of a gamma process $X(t)$ with parameters β and $R(\tau)$. It is assumed that $R(0) = a = 1$, the moments for the more general case with arbitrary a can be obtained by multiplication with powers of a since $aX(t)$ has parameters β and $aR(\tau)$. (The power of a to be used is the sum of the indices of the given moment.) The moments belong to the joint distribution of $X(t_1)$, $X(t_2)$, $X(t_3)$, etc. and we use the following notation:

$$\beta(\beta + 1) \dots (\beta + i) = \beta_i \quad (i = 1, 2, \dots);$$

$$R(|t_i - t_j|) = \sigma_{ij} = \sigma_{ji}.$$

$$\mu'_1 = \beta, \quad \mu'_i = \beta_{i-1} \quad (i = 2, \dots)$$

$$\mu'_{11} = \beta_1 - \beta(1 - \sigma_{12}^2)$$

$$\mu'_{21} = \beta_2 - 2\beta_1(1 - \sigma_{12}^2)$$

$$\mu'_{22} = \beta_3 - 4\beta_2(1 - \sigma_{12}^2) + 2\beta_1(1 - \sigma_{12}^2)^2$$

$$\mu'_{31} = \beta_3 - 3\beta_2(1 - \sigma_{12}^2)$$

$$\begin{aligned} \mu'_{111} = & \beta_2 - \beta_1(3 - \sigma_{12}^2 - \sigma_{13}^2 - \sigma_{23}^2) \\ & + \beta(1 - \sigma_{12}^2 - \sigma_{13}^2 - \sigma_{23}^2 + 2\sigma_{12}\sigma_{23}\sigma_{13}) \end{aligned}$$

$$\begin{aligned} \mu'_{121} = & \beta_3 - \beta_2(5 - 2\sigma_{12}^2 - \sigma_{13}^2 - 2\sigma_{23}^2) \\ & + 2\beta_1(2 - 2\sigma_{12}^2\sigma_{13}^2 - 2\sigma_{23}^2 + \sigma_{12}^2\sigma_{23}^2 + 2\sigma_{12}\sigma_{13}\sigma_{23}) \\ \mu'_{1111} = & \beta^4 + \beta^3(\sigma_{12}^2 + \sigma_{13}^2 + \sigma_{14}^2 + \sigma_{23}^2 + \sigma_{24}^2 + \sigma_{34}^2) \end{aligned}$$

$$\begin{aligned} & + \beta^2(2(\sigma_{12}\sigma_{13}\sigma_{23} + \sigma_{12}\sigma_{14}\sigma_{24} + \sigma_{13}\sigma_{34}\sigma_{14} \\ & + \sigma_{23}\sigma_{34}\sigma_{24}) + \sigma_{12}^2\sigma_{34}^2 + \sigma_{13}^2\sigma_{24}^2 + \sigma_{14}^2\sigma_{23}^2) \\ & + 2\beta(\sigma_{12}\sigma_{23}\sigma_{34}\sigma_{14} + \sigma_{13}\sigma_{34}\sigma_{24}\sigma_{12} + \sigma_{14}\sigma_{24}\sigma_{23}\sigma_{13}) \end{aligned}$$

$$\mu_2 = \beta$$

$$\mu_3 = 2\beta$$

$$\mu_4 = 3\beta(\beta + 2)$$

$$\mu_{11} = \beta\sigma_{12}^2$$

$$\mu_{21} = 2\beta\sigma_{12}^2$$

$$\mu_{22} = \beta(\beta(1 + 2\sigma_{12}^4) + 2\sigma_{12}^2(2 + \sigma_{12}^2))$$

$$\mu_{31} = 3\beta(\beta + 2)\sigma_{12}^2$$

$$\mu_{111} = 2\beta\sigma_{12}\sigma_{13}\sigma_{23}$$

$$\mu_{121} = \beta(\beta(2\sigma_{12}^2\sigma_{23}^2 + \sigma_{13}^2) + 2(2\sigma_{12}\sigma_{13}\sigma_{23} + \sigma_{12}^2\sigma_{23}^2))$$

$$\mu_{1111} = \beta^2(\sigma_{12}^2\sigma_{34}^2 + \sigma_{13}^2\sigma_{24}^2 + \sigma_{14}^2\sigma_{23}^2)$$

$$+ 2\beta(\sigma_{12}\sigma_{23}\sigma_{34}\sigma_{14} + \sigma_{13}\sigma_{34}\sigma_{24}\sigma_{12} + \sigma_{14}\sigma_{24}\sigma_{23}\sigma_{13})$$

Some of these moments were used to calculate variances and covariances of estimators based on observations from a particular gamma process. (Cf. Section 5.2.)

4. The gamma point process

4.1. *The process of distances.* From the gamma process $X(t)$, $t \in R_1$, with parameters β and $R(\tau)$, we can define a stationary discrete parameter process

$$X_i = X(i), \quad i \in I,$$

The process X_i , $i \in I$, is non-negative and stationary, it can be regarded as the process of *distances* between points of a stationary point process. We call the corresponding point process the *gamma point process* with parameters β and $R(r)$, $r = 0, 1, 2, \dots$

X_i has mean $a\beta$, where $a = R(0)$, and covariance function $\beta R^2(|r|)$. The intensity of the process, we shall denote it by ν , is given by

$$\nu = \frac{1}{E(X_i)} = \frac{1}{a\beta}$$

4.2. *The process of lengths.* As mentioned before, in collecting data on point processes one usually decides on an arbitrary time t and observes the points which occurred after t . In such a case one does not observe a part of a realisation of the process of distances, X_i , but a part of a realisation of the process of lengths, L_i .

We call L_0 the length of the interval in progress at t and the following lengths $L_1, L_2, \dots$,

Many of the results of McFadden (1962) can be obtained by the following heuristic argument (see also Cox and Lewis (1966) p. 61).

The probability to select a L_0 in the interval $(x, x + dx)$ is proportional to x and to $f_{X_i}(x) dx$, thus

$$f_{L_0}(x) dx \propto x f_{X_i}(x) dx$$

$$= \frac{1}{E(X_i)} x f_{X_i}(x) dx = \nu x f_{X_i}(x) dx.$$

The conditional probability density function of $L_1, L_2, \dots, L_n$, given that $L_0 = x_0$, is identical to the conditional probability density function of $X_{i+1}, \dots, X_{i+n}$, given that $X_i = x_0$. (Both depend on the extent x_0 of the first of $n+1$ intervals, it is entirely irrelevant how this first interval was obtained, i.e. whether x_0 is a *length* or a *distance*.) Therefore

$$\begin{aligned} & f_{L_0, L_1, \dots, L_n}(x_0, x_1, \dots, x_n) \\ &= f_{L_1, L_2, \dots, L_n | L_0}(x_1, x_2, \dots, x_n | x_0) f_{L_0}(x_0) \\ &= f_{X_{i+1}, X_{i+2}, \dots, X_{i+n} | X_i}(x_1, x_2, \dots, x_n | x_0) \nu_{x_0} f_{X_i}(x_0) \\ &= \nu_{x_0} f_{X_i, X_{i+1}, \dots, X_{i+n}}(x_0, x_1, \dots, x_n). \end{aligned}$$

The p.d.f. of the marginal distributions of L_0 and any subset of the $L_1, L_2, \dots, L_n$, can be obtained by integrating over some of the variables. For example

$$\begin{aligned} & f_{L_0, L_r, \dots, L_{r+m}}(x_0, x_r, \dots, x_{r+m}) = \\ &= \nu_{x_0} f_{X_i, X_{i+r}, \dots, X_{i+r+m}}(x_0, x_r, \dots, x_{r+m}) \quad (0 < r \leq n-m; 0 \leq m < n). \end{aligned}$$

The relationship between moments (as far as they exist) of the joint distribution of lengths and the joint distribution of distances are then obvious:

$$E(L_0^j L_1^k \dots L_n^s) = \nu E(X_i^{j+1} X_{i+2}^k \dots X_{i+n}^s)$$

One can also obtain the ch.f. of $L_0, L_1, \dots, L_n$ from that of $X_i, X_{i+1}, \dots, X_{i+n}$:

$$\begin{aligned} \varphi_{L_0, L_1, \dots, L_n}(u_0, u_1, \dots, u_n) &= \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f e^{i(u_0 x_0 + \dots + u_n x_n)} \\ &\quad \nu_{x_0} f_{X_i, X_{i+1}, \dots, X_{i+n}}(x_0, x_1, \dots, x_n) dx_0 \dots dx_n \\ &= \frac{\nu}{i} \frac{\partial}{\partial u_0} (\varphi_{X_i, X_{i+1}, \dots, X_{i+n}}(u_0, u_1, \dots, u_n)) \end{aligned}$$

For the gamma point process the ch.f. of the lengths $L_1, L_2, \dots, L_n$ is then

$$\begin{aligned} \varphi_{L_1, L_2, \dots, L_n}(u_1, u_2, \dots, u_n) &= \frac{\nu}{i} \left(\frac{\partial}{\partial u_0} |1 - i D(u_0, u_1, \dots, u_n) \sum_{n+1}^{-\beta}| u_0 = 0 \right) \\ &= \frac{-1}{i a} |1 - i D(0, u_1, u_2, \dots, u_n) \sum_{n+1}^{-(\beta+1)}| \\ &\quad |D(0, 1, \dots, 1) - i D(1, u_1, u_2, \dots, u_n) \sum_{n+1}| \end{aligned}$$

In the sequel we shall need

$$E(L_r) \text{ and } \text{Cov}(L_r, L_{r+m}) \quad (r=1, 2, \dots, n; m=0, 1, \dots, n-r).$$

From the equation relating the moments of the lengths to the moments of the distances one has

$$E(L_r) = \nu E(X_i X_{i+r})$$

and

$$E(L_r L_{r+m}) = \nu E(X_i X_{i+r} X_{i+r+m}).$$

For the gamma point process with parameters β and $R(r)$, $r \geq 0$ integer, one obtains (where $a = R(0)$)

$$E(L_r) = R^2(r)/a + a\beta$$

$$\text{Cov}(L_r L_{r+m}) = \beta R^2(m) + 2 R(r) R(m) R(r+m)/a - R^2(r) R^2(r+m)/a^2$$

4.3 The waiting times

The sum of n distances is usually called the waiting time to the n th point:

$$W_n = \sum_{k=1}^n X_{j+k}.$$

Its characteristic function is

$$\varphi_{W_n}(u) = \varphi_{X_{j+1}, \dots, X_{j+n}}(u, u, \dots, u) = |I - i u \Sigma_n|^{-\beta}$$

Because for a $n \times n$ matrix A of rank n with characteristic values κ_j , $j=1, 2, \dots, n$,

$$|A - \kappa I| = \prod_{j=1}^n (\kappa_j - \kappa),$$

$$\varphi_{W_n}(u) = \prod_{j=1}^n (1 - i u \lambda_j)^{-\beta},$$

where the λ_j ($j=1, 2, \dots, n$) are the characteristic values of Σ_n . W_n is thus distributed as the sum of n independent gamma distributed random variables, where the j -th variable has parameters λ_j and β ($j=1, 2, \dots, n$).

Since the multivariate gamma distribution is, for $\beta = k/2$, k integer, identical to the joint distribution of the diagonal elements of a Wishart matrix, W_n is, for $\beta = k/2$, distributed as the trace of a Wishart matrix with parameters k and $(1/2) \Sigma_n$. ($W(\Sigma_n/2, k)$).

It is also easily shown that the quadratic form $x'Mx$ is, if x is $N(0, V)$, distributed as the trace of a matrix which has a $W(MV, 1)$ distribution. The distribution of the waiting times W_n is therefore closely related to the distribution of quadratic forms in normal variables.

It follows from the last two paragraphs that we can use results obtained in multivariate normal analysis, in the study of quadratic forms and the study of Wishart's distribution.

Troskie (1969) has obtained an expression for the exact density of trace A , where A is $W(\Sigma_p, n)$, which is valid for $|z| < 1$:

$$f_{\text{tr}A}(z) = (z^{np/2-1}/2^{np/2} \Gamma(np/2) |\Sigma_p|^{n/2}) \cdot$$

$$\bullet \sum_{k=0}^{\infty} \frac{z^k}{(np/2)_{\kappa}} \sum_{\kappa} (n/2)_{\kappa} C_{\kappa}(-\hat{\Sigma}_p^{-1}/2).$$

The second sum is over all partitions κ of k :

$$\kappa = (k_1, k_2, \dots, k_p) \text{ with } k_1 \geq k_2 \geq \dots \geq k_p \geq 0$$

and $\sum k_i = k$. $C_{\kappa}(S)$ is the zonal polynomial of S belonging to κ . (Cf. e.g. James, 1964) and

$$(a)_{\kappa} = \prod_{i=1}^p (a - (i-1)/2)_{k_i}, \text{ with } (a)_{\kappa} = \Gamma(a + k) / \Gamma(a).$$

This result was obtained from the joint density of the characteristic values of A (James, 1964), using an integral by Khatri and Pillai (1968).

There is also considerable literature on the distribution of quadratic forms in normal variables, a recent paper in which further references to previous work are given is that by Kotz, Johnson and Boyd (1967).

The exact result quoted above and most of the expressions and approximations for the distribution of quadratic forms are too complicated for our purposes. A simple and obvious approximation to the distribution of W_n should be sufficient: one may use a gamma distribution which has the same mean and variance as the distribution of W_n . Box (1954) studied this approximation, which he attributed to Satterthwaite (1941), and reported a reasonably good fit.

From the ch.f. of W_n one gets

$$E(W_n) = \beta \sum \lambda_j$$

$$\text{Var}(W_n) = \beta \sum \lambda_j^2$$

Now $\sum \lambda_j = \text{trace}(\Sigma_n) = n a$ and

$$\begin{aligned} \sum \lambda_j^2 &= \text{trace}(\Sigma_n^2) = n R^2(o) + 2 \sum_{j=1}^{n-1} (n-j) R^2(j) \\ &= n a^2 + 2 n a^2 \gamma(n), \text{ where} \end{aligned}$$

$$\gamma(n) = \sum_{j=1}^{n-1} (1 - j/n) R^2(j) / a^2$$

Thus

$$E(W_n) = n a \beta$$

$$\text{Var}(W_n) = n a^2 \beta (1 + 2 \gamma(n))$$

We have therefore:

W_n is approximately distributed as a gamma variable with parameters $a^* = a(1 + 2 \gamma(n))$ and $\beta^* = n \beta / (1 + 2 \gamma(n))$.

Another random variable, analogous to W_n , but derived from the *lengths* is also of interest. It is the length of the interval from the time at which the observation of the process started to the n th observed point, we denote it by V_n . If the length of the interval between the time at which observations started and the first point (the "forward recurrence time") is denoted by W , then

$$V_1 = W$$

$$V_n = W + L_1 + L_2 + \dots + L_{n-1} \quad \text{for } n = 2, 3, \dots$$

Given $L_0 = x_0$, the distribution of W is uniform over $(0, x_0)$.

Also

$$\begin{aligned} f_{W|L_0, L_1, \dots, L_n}(x_{-1}|x_0, x_1, \dots, x_n) \\ = f_{W|L_0}(x_{-1}|x_0) = \frac{1}{x_0} \end{aligned} \quad \begin{aligned} 0 \leq x_{-1} \leq x_0 \\ n = 0, 1, \dots \end{aligned}$$

Therefore

$$\begin{aligned} f_{W, L_0, \dots, L_n}(x_{-1}, x_0, \dots, x_n) \\ = f_{W|L_0, \dots, L_n}(x_{-1}|x_0, \dots, x_n) f_{L_0, \dots, L_n}(x_0, \dots, x_n) \\ = \nu f_{X_j, \dots, X_{j+n}}(x_0, x_1, \dots, x_n) \end{aligned} \quad \begin{aligned} x_{-1} \leq x_0 \leq \infty \\ 0 \leq x_j \leq \infty \\ (j=1, 2, \dots, n) \end{aligned}$$

The joint density of $W, L_1, \dots, L_n$ is thus

$$\begin{aligned} f_{W, L_1, \dots, L_n}(x_{-1}, x_1, \dots, x_n) \\ = \nu \int_{x_{-1}}^{\infty} f_{X_j, X_{j+1}, \dots, X_{j+n}}(x_0, x_1, \dots, x_n) dx_0 \end{aligned} \quad n = 1, 2, \dots$$

and the marginal density of W

$$f_W(x) = \int_x^{\infty} \frac{1}{x_0} f_{L_0}(x_0) dx_0 = \nu \int_x^{\infty} f_{X_j}(t) dt$$

Using integration by parts one can then obtain the following identity for the ch.f. of $W, L_1, \dots, L_n$

$$\begin{aligned} \varphi_{W, L_1, \dots, L_n}(u_{-1}, u_1, \dots, u_n) \\ = \frac{\nu}{i u_{-1}} (\varphi_{X_j, X_{j+1}, \dots, X_{j+n}}(u_{-1}, u_1, \dots, u_n) \\ - \varphi_{X_{j+1}, X_{j+2}, \dots, X_{j+n}}(u_1, u_2, \dots, u_n)) \end{aligned} \quad n=1, 2, \dots$$

One gets thus the following identity for the ch.f.'s of V_n and W_n :

$$\varphi_{V_n}(u) = \frac{\nu}{iu} (\varphi_{W_n}(u) - \varphi_{W_{n-1}}(u)), \quad n=1,2,\dots,$$

where we read

$$\varphi_{W_0}(u) \equiv 1.$$

For the gamma point process one gets

$$\varphi_{W_{L_1, \dots, L_n}}(u_1, u_1, \dots, u_n) = \frac{1}{i u_1 \alpha \beta} \cdot \\ (|1 - iD(u_1, u_1, \dots, u_n) \Sigma_{n+1}|^{-\beta} - |1 - iD(u_1, u_2, \dots, u_n) \Sigma_n|^{-\beta})$$

From this follows straightaway that the ch.f. of V_n is given by

$$\varphi_{V_n}(u) = \frac{1}{i u \alpha \beta} (|1 - iu \Sigma_n|^{-\beta} - |1 - iu \Sigma_{n-1}|^{-\beta}) \quad n=2,3,\dots \\ \varphi_{V_1}(u) = \varphi_W(u) = \frac{1}{i u \alpha \beta} (|1 - iu \Sigma_1|^{-\beta} - 1).$$

This ch.f. could also be expressed in terms of the characteristic values of Σ_n and Σ_{n-1} (as it was done above for the ch.f. of W_n). A gamma distribution should also be a reasonably good approximation for the distribution of V_n .

4.4 The counting processes and the covariance density

One must here distinguish two counting processes: one belongs to the lengths and is denoted by $N(\tau)$ and the other belongs to the distances, and is denoted by $M(\tau)$.

$N(\tau)$ is the number of points in the interval $(t, t+\tau]$ where t is the (arbitrary) time at which the observations started. $M(\tau)$ is the number of points occurring in the interval $(t_p, t_p+\tau]$ where at t_p a point has occurred and the observation of the process has started at t_p .

It is obvious that

$$P(N(\tau) < n) = P(V_n > \tau) \quad n=1,2,\dots$$

and therefore

$$P(N(\tau) = n) = F_{V_n}(\tau) - F_{V_{n+1}}(\tau) \quad n=0,1,\dots$$

and that

$$P(M(\tau) < n) = P(W_n > \tau) \quad n=1,2,\dots$$

and therefore ,

$$P(M(\tau) = n) = F_{W_n}(\tau) - F_{W_{n+1}}(\tau) \quad n=0,1,\dots,$$

where one has to read $F_{V_0}(\tau) \equiv F_{W_0}(\tau) \equiv 1$.

The above probabilities could be calculated approximately using the gamma approximation for the distribution of W_n (or V_n) together with tables of the incomplete gamma function. The convenient relationship between the distribution function of a gamma variable with integer parameter β and the Poisson distribution function:

$$\int_0^x \frac{x^{\beta-1} e^{-x/a}}{a \beta \Gamma(\beta)} = \sum_{j=0}^{\beta-1} (\lambda/a)^j e^{-\lambda/a} / j!$$

(cf. e.g. Cox and Lewis (1966) p. 24),

which makes the calculation of the mentioned probabilities particularly easy in the case of a Poisson or an Erlang process, breaks down if β is not integer. It may nevertheless be sometimes advisable to use it, and to interpolate.

We shall now obtain an identity involving the ch.f.s of $N(\tau)$ and $M(\tau)$, and the exact first two moments of $N(\tau)$.

From the above one has

$$\varphi_{N(\tau)}(u; \tau) = \sum_{n=0}^{\infty} e^{iun} (F_{V_n}(\tau) - F_{V_{n+1}}(\tau))$$

$$\varphi_{M(\tau)}(u; \tau) = \sum_{n=0}^{\infty} e^{iun} (F_{W_n}(\tau) - F_{W_{n+1}}(\tau))$$

Taking Laplace transforms w.r.t. τ results in

$$s \varphi_{N(\tau)}^*(u; s) = \sum_{n=0}^{\infty} e^{iun} (f_{V_n}^*(s) - f_{V_{n+1}}^*(s))$$

$$s \varphi_{M(\tau)}^*(u; s) = \sum_{n=0}^{\infty} e^{iun} (f_{W_n}^*(s) - f_{W_{n+1}}^*(s)) \quad (f_{V_0}^*(s) \equiv f_{W_0}^*(s) \equiv 1)$$

From the relationship between the ch.f.s of V_n and W_n one finds

$$f_{V_n}^*(s) = \frac{\nu}{s} (f_{W_{n-1}}^*(s) - f_{W_n}^*(s)) \quad n=1, 2, \dots$$

Thus

$$s \varphi_{N(\tau)}^*(u; s) = 1 + (e^{iu} - 1) \sum_{n=1}^{\infty} e^{iun(n-1)} f_{V_n}^*(s)$$

and

$$\sum_{n=1}^{\infty} e^{iun(n-1)} f_{V_n}^*(s)$$

$$= \sum_{n=0}^{\infty} e^{iun} f_{V_{n+1}}^*(s) = \frac{\nu}{s} \sum_{n=0}^{\infty} e^{iun} (f_{W_n}^*(s) - f_{W_{n+1}}^*(s))$$

$$= \nu \varphi_{M(\tau)}^*(u; s)$$

Therefore one gets the following identity:

$$\varphi_{N(\tau)}^*(u; s) = \frac{1}{s} (1 + (e^{iu} - 1) \nu \varphi_{M(\tau)}^*(u; s))$$

and after taking inverse Laplace transforms

$$\varphi_{N(\tau)}(u; \tau) = 1 + \nu (e^{iu} - 1) \int_0^\tau \varphi_{M(\tau)}(u; x) dx.$$

In terms of moment functions this can be written as

$$\mu'_{h-1}(N(\tau)) = \nu \sum_{j=1}^h \binom{h}{j} \int_0^\tau \mu'_{h-j}(M(x)) dx$$

In particular one gets

$$\mu'_1(N(\tau)) = \nu \tau$$

$$\mu'_2(N(\tau)) = \nu (\tau + 2 \int_0^\tau H(x) dx)$$

$$\text{Var}(N(\tau)) = \nu (\tau + 2 \int_0^\tau (H(x) - \nu x) dx).$$

The coefficient of dispersion $\text{Var}(N(\tau))/E(N(\tau))$, which is one for a Poisson process $(H(x) \equiv \nu x)$, can thus be *larger* or *smaller* than one, depending on $H(\tau)$.

The function $H(\tau) = E(M(\tau)) = \mu'_1(M(\tau))$, the *mean value function* (or *renewal function*) is a very important parameter function of the point process. Giesekus (1950) has introduced the function

$$\psi(\tau) = H'(|\tau|) - \nu,$$

the derivative of the difference between the mean value function of the point process and the mean value function of a Poisson process with the same intensity ν and called it the *statistical anomaly*.

The function $\nu \psi(\tau)$, which Bartlett (1955) calls the *covariance density*, can be regarded as the covariance function of the point process.

For the gamma point process

$$\varphi^*_{M(\tau)}(u; s) = \frac{1}{s} (1 + (e^{iu} - 1) \sum_{n=1}^{\infty} e^{iu(n-1)} |1 + s \sum_n |^{-\beta})$$

and

$$\begin{aligned} \varphi^*_{N(\tau)}(u; s) \\ = \frac{1}{s} (1 + (e^{iu} - 1) \frac{\nu}{s} (1 + (e^{iu} - 1) \sum_{n=1}^{\infty} e^{iu(n-1)} |1 + s \sum_n |^{-\beta})) \end{aligned}$$

The Laplace transform of the mean value function is

$$\begin{aligned} H^*(s) &= \frac{1}{s} \sum_{n=1}^{\infty} |1 + s \sum_n |^{-\beta} \\ &= \frac{1}{s} h^*(s), \end{aligned}$$

where $h^*(s)$ is the Laplace transform of $H'(\tau) = h(\tau)$, say.

One obviously has also

$$h(\tau) = \sum_{n=1}^{\infty} f_{W_n}(\tau), \quad H(\tau) = \sum_{n=1}^{\infty} F_{W_n}(\tau),$$

and one could derive approximations using the gamma approximation for the distribution of W_n which was discussed above.

The Laplace transform of the h -th moment of $M(\tau)$ is

$$\frac{1}{s} \sum_{n=1}^{\infty} (n^h \cdot (n-1)^h) |1 + s \Sigma_n|^{-\beta}.$$

4.5 The process of numbers of points in adjacent intervals of equal length

There are many more questions which one could ask about the counting processes, in particular about $N(\tau)$, which is usually the process on which observations are available. $N(\tau)$ is a stochastic process which has as index set all pairs (t, τ) , $-\infty \leq t \leq +\infty$, $\tau \geq 0$. To indicate the dependence on t we write in this section $N_t(\tau)$ for the number of points in $(t, t+\tau]$. For fixed τ this process is stationary with respect to t , Gieseke (1950) gave it covariance function

$$\text{Cov}(N_t(\tau), N_{t+v}(\tau))$$

$$= \nu(\tau - |v|) + \int_{-\tau}^{\tau} \psi(v-x)(\tau - |x|) dx \quad \text{if } |v| \leq \tau$$

$$= \nu \int_{-\tau}^{\tau} \psi(v-x)(\tau - |x|) dx \quad \text{if } \tau \leq |v|$$

Of particular interest is sometimes the discrete parameter process of numbers of points in adjacent intervals of constant length:

$$Y_i = N_{t+i\tau}(\tau) \quad i=0,1,\dots$$

From the covariance function of the increments one may obtain the covariance function of Y_i

$$\text{Cov}(Y_i, Y_{i+r}) = \nu \int_{-\tau}^{\tau} \psi(r\tau - x)(\tau - |x|) dx \quad r=1,2,\dots$$

which can, in terms of the mean value function, be written as

$$\text{Cov}(Y_i, Y_{i+r}) = \nu \left(\int_{r\tau}^{(r+1)\tau} H(u) du - \int_{(r-1)\tau}^{r\tau} H(u) du \right) - \nu^2 \tau^2.$$

5. Problems of estimation

5.1 The gamma point process with exponential covariance

At this stage one has to choose a function $R^*(x,y) = R(|x-y|)$ which is real valued and positive definite and which generates the covariance function of the process of distances. The choice is, of course, very wide. There may be cases where the way in which the data are generated may give rise to a very particular function R . (The process of distances may have originated by squaring of a Gaussian process, and the form of the covariance function of the Gaussian process may be known.) But usually one will have to guess, possibly guided by an estimated covariance function.

A very simple possibility, which we shall discuss in detail here, is to assume that

$$R^*(x,y) = a e^{-\lambda |x-y|}, \quad 0 < a, \lambda < \infty$$

thus

$$R(\tau) = a e^{-\lambda \tau}$$

This leads to a process of distances with mean $a\beta$ and with covariance function $a^2 \beta e^{-2\lambda |r|}$. It is now convenient to replace the three parameters a, β and λ by μ, σ^2 and ρ , where

$$\begin{aligned} \mu &= a\beta \\ \sigma^2 &= a^2 \beta \\ \rho &= e^{-2\lambda} \end{aligned}$$

We are then dealing with a process of distances with mean μ , variance σ^2 and covariance function $\sigma^2 \rho^{|r|}$. The corresponding point process will be called a gamma point process with exponential covariance.

The problem which has to be solved is the estimation of the parameters μ, σ^2 and ρ .

The data available are usually *lengths* and not *distances*. We shall at first search for methods which are suitable for *distances*, and then consider how these methods have to be modified to be suitable for *lengths*.

5.2 Estimation on the basis of distances

The method of maximum likelihood cannot be used because of the prohibitive nature of the joint density function of the distances. One could, however, use the asymptotic maximum likelihood method (Whittle (1952)).

Whittle's method (cf. also Hannan (1960) p. 46) is an asymptotic maximum likelihood method for Gaussian processes, but can also be regarded as a method of least squares for arbitrary stationary processes. The method is limited (whether the process is Gaussian or not) to the case in which the covariance generating function of the process,

$$\Lambda(z) = \sum_{r=-\infty}^{+\infty} \text{Cov}(X_i, X_{i+r}) z^r$$

is such that $\Lambda(z)$ and $(\Lambda(z))^{-1}$ exist for $|z| = 1$. In our case.

$$\Lambda(z) = \sigma^2 (1 - \rho^2) / (1 - \rho z)(1 - \rho/z)$$

satisfies this condition.

The method has been derived for processes with zero mean, "but the degree of approximation is such that corrections for the mean do not affect" the method (Whittle (1952) p. 424). In our case corrections for the mean have to be made.

The mean μ must obviously be estimated by $\sum_{i=1}^n X_i/n$ if n distances are available.

One must then minimise, w.r.t. ρ , the integral

$$\frac{1}{2\pi i} \int_C \frac{\Lambda(z) z^r}{\Lambda(z)} \frac{dz}{z}$$

where c is the unit circle and

$$\log A = \frac{1}{2\pi i} \int_c \log \Lambda(z) \frac{dz}{z}$$

(Whittle (1952) formulae (1.12) and (1.4)), and

$$C_r = \frac{1}{n-r} \sum_{i=1}^{n-r} (X_i - \bar{X}_r') (X_{i+r} - \bar{X}_r'')$$

where

$$\bar{X}_r' = \frac{1}{n-r} \sum_{i=1}^{n-r} X_i, \quad \bar{X}_r'' = \frac{1}{n-r} \sum_{i=1}^{n-r} X_{i+r}$$

In our case $A = \sigma^2(1-\rho^2)$,

therefore $\Lambda(z)/A = 1/(1-\rho z)(1-\rho/z)$. The integral is then easily evaluated and becomes

$$C_0(1 + \rho^2) - 2C_1\rho.$$

It is minimised if $\hat{\rho} = C_1/C_0$. The minimum value provides an estimator of A , and therefore of σ^2 :

$$\hat{A} = C_0(1 + C_1^2/C_0^2) - 2C_1^2/C_0 = C_0(1 - \hat{\rho}^2), \quad \hat{\sigma}^2 = C_0$$

We obtain now the standard errors of the estimators. The variance of $\sum_{i=1}^n X_i/n$ is

$$\frac{1}{n} \sum_{r=-n+1}^{n-1} (1-|r|/n) \text{Cov}(X_i, X_{i+r})$$

(cf. e.g. Cox and Lewis (1966) p. 87) which in our case becomes

$$\frac{\sigma^2}{n} \left(\frac{1+\rho}{1-\rho} + (2\rho/n) \frac{\rho^{n-1}}{(\rho-1)^2} \right)$$

This can for large n be approximated by

$$\frac{\sigma^2}{n} \left(\frac{1+\rho}{1-\rho} \right).$$

The standard error of the estimator of μ is therefore

$$\left(\frac{C_0(C_0 + C_1)}{n(C_0 - C_1)} \right)^{1/2}$$

The moments given in Section 3 were used to obtain approximations to the variances of C_0 and C_1 and to their covariance. For that the given moments had to be specialised for the case of a gamma process with parameter function $R(\tau) = a e^{-\lambda|\tau|}$. (The given moments have to be multiplied by a^r , where r is the sum of the indices of the moment, and $\sigma_{ij}^2 = e^{-2\lambda|t_i - t_j|} = \rho^{|t_i - t_j|}$).

Terms of order $1/n^2$ and terms of order ρ^n were neglected. We obtained

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 \right) \approx \frac{2\sigma^4}{n(1-\rho^2)} (1 + \rho^2 + \frac{\sigma^2}{\mu^2} (3 + 4\rho + 3\rho^2))$$

$$\text{Var} \left(\frac{1}{n-1} \sum_{i=1}^{n-1} (X_i - \mu)(X_{i+1} - \mu) \right) \approx$$

$$\frac{\sigma^4}{n(1-\rho^2)} (1+4\rho^2 - \rho^4 + \frac{2\rho\sigma^2}{\mu^2} (2+7\rho+2\rho^2 \cdot \rho^3))$$

$$\text{Cov}(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2, \frac{1}{n-1} \sum_{i=1}^{n-1} (X_{i+1} - \mu)) \approx$$

$$\frac{4\sigma^4\rho}{n(1-\rho^2)} (1 + \frac{\sigma^2}{\mu^2} (3+2\rho))$$

The first result can be used to find a standard error of C_0 , if the variation in the correction for the mean is neglected. The second and third, together with

$$E(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2) = \sigma^2$$

and

$$E(\frac{1}{n-1} \sum_{i=1}^{n-1} (X_i - \mu)(X_{i+1} - \mu)) = \sigma^2 \rho$$

can be used to provide an approximate variance of $\hat{\rho} = C_1/C_0$, since

$$\text{Var}(C_1/C_0) \approx (1/E^2(C_0))$$

$$(\text{Var}(C_1) - 2 E(C_1) \text{Cov}(C_1, C_0)/E(C_0) + E^2(C_1) \text{Var}(C_0)/E^2(C_0)).$$

One gets

$$\text{Var}(\frac{1}{n-1} \sum_{i=1}^{n-1} (X_i - \mu)(X_{i+1} - \mu) / \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2)$$

$$\approx \frac{1}{n(1-\rho^2)} (1-8\rho+6\rho^2+\rho^4 + \frac{4\rho\sigma^2}{\mu^2} (1-\rho-\rho^2 + \rho^3)).$$

This last result gives, if the variation of the correction for the mean is neglected, a standard error of $\hat{\rho}$. (To obtain standard errors, μ , σ^2 and ρ have to be replaced by their estimators $\bar{X}$, C_0 , and $\hat{\rho} = C_1/C_0$.)

The estimator of μ is unbiased, the estimators of σ^2 and ρ have a small bias which vanishes for large n .

5.3 Estimation on the basis of lengths

As we mentioned before, distances are very rarely observed, one usually has observations on the process of lengths. We shall show that $E(L_m)$ and $\text{Cov}(L_m, L_{m+r})$ approach quickly, for increasing m , $E(X_j)$ and $\text{Cov}(X_j, X_{j+r})$ respectively. Because of this, since all estimation procedures suggested in Section 5.2 involve only estimators of moments of order up to two, we shall suggest to omit the first few of the observed lengths and to use the remaining observations as if they were observed distances.

From the results obtained at the end of Section 4.2 one gets

$$E(L_m) = \mu (1 + \frac{\sigma^2}{\mu^2} \rho^m)$$

$$\text{Cov}(L_m, L_{m+r}) = \sigma^2 \rho^r (1 + \frac{\sigma^2}{\mu^2} \rho^m (2 - \rho^m))$$

$$\begin{matrix} m=1,2,\dots \\ r=0,1,\dots \end{matrix}$$

The relative difference between the covariance functions,

$$(\text{Cov}(L_m, L_{m+r}) - \text{Cov}(X_i, X_{i+r})) / \text{Cov}(L_m, L_{m+r})$$

is independent of r , it is smaller than ϵ if and only if

$$2(\sigma^2 / \mu^2) \rho^m < \epsilon / (1 - \epsilon).$$

(The term of order ρ^{2m} is neglected.) Apart from the squared coefficient of variation, σ^2 / μ^2 , the error made, if lengths are treated as if they were distances, depends on ρ^m : it is small if ρ^m is small. This is what one would expect, since ρ^m is the correlation function at lag m .

It is not necessary to consider the relative difference between the means, since it is always smaller than that between the covariances.

To give an example: if $\rho = 1/2$ and $\sigma^2 / \mu^2 = 1$ the relative differences are smaller than .01 if m is larger than seven.

Unless ρ is near to one, omitting the first ten or so observations should in most cases ensure that the methods for distances can be employed for lengths.

On point processes one usually observes many lengths and omitting a few will not hurt. If every observation is precious and all observations have to be used in order to ensure tolerable standard errors one has to study the introduced bias. For the mean it is easily obtained:

$$E\left(\frac{1}{n} \sum_{i=1}^n L_i\right) = \mu \left(1 + \frac{\sigma^2 \rho (\rho^n - 1)}{n \mu^2 (\rho - 1)}\right)$$

For the estimators of σ^2 and ρ the necessary algebra is very laborious, it was not carried out, also in view of the fact that the estimators described in Section 5.2 are only asymptotically unbiased.

The following argument may, however, help to see the problem in its proper perspective. Call the smallest value of r for which $\rho_r = \rho^{|r|}$ is „negligible“ (say smaller than .01) r_0 . The estimators of μ, σ^2, ρ_r are — if calculated from *lengths* — in fact *averages of estimators* of $E(L_m)$, $\text{Var}(L_m)$ and $\text{Cov}(L_m, L_{m+r})$, for $m = 1, 2, \dots, n$ (or $1, 2, \dots, n-r$ in the case of the covariance). Only for $m < r_0$ are $E(L_m)$, $\text{Var}(L_m)$, $\text{Cov}(L_m, L_{m+r})$ noticeably different from the parameters which one actually wants to estimate: $\mu = E(X_i)$, $\sigma^2 = \text{Var}(X_i)$, $\sigma^2 \rho_r = \text{Cov}(X_i, X_{i+r})$. (This is intuitively plausible and is also borne out by the above given formulae.) If r_0 is small compared with n (or $n-r$) the influence of the first r_0 terms on the average over n (or $n-r$) terms is small.

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Manuscript received 3rd July 1969.